

Analysis of Boeing 767-300 Airliner

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This paper examines the design characteristics of the Boeing 767-300 airliner, focusing on its aerodynamic and stability properties. A geometric recreation of the main wing and tail parameters is conducted, along with center of gravity calculations for various payloads and flight conditions. DATCOM is used to determine stability derivatives in the longitudinal and lateral directions, as well as to analyze the neutral point, static margin, and phugoid characteristics. Additionally, the landing gear design is evaluated. This study provides insight into the engineering principles that have contributed to the Boeing 767-300's effectiveness and longevity in commercial aviation.

I. Nomenclature

C_o	= Root Chord
C_T	= Tip Chord
$\bar{c}$	= Mean Aerodynamic Chord
c	= chord
S	= Wing Area
b	= Wing Span
Δ	= Hinge Line
λ	= Taper Ratio
$X_{cg Forward}$	= x-location of Center of Gravity in Forward Conditions
$X_{cg Aft}$	= x-location of Center of Gravity in Aft Conditions
Z_{cg}	= z-location of Center of Gravity
Θ	= Tip back Angle
Φ	= Turnover Angle

II. Introduction

The Boeing 767-300 is a mid- to long-range, wide-body twin-engine jet airliner developed by Boeing in the early 1980s as part of the 767 family. Designed for increased passenger capacity and extended range, the 767-300 features an elongated fuselage compared to its predecessor, the 767-200, while maintaining fuel efficiency and operational flexibility. Since its first flight in 1986 and subsequent entry into service in 1988, the aircraft has been widely adopted by commercial airlines and cargo operators due to its reliable performance, advanced aerodynamics, and efficient twin-engine configuration.

III. Geometric Recreation of Main Wing and Tail

In order to get a geometric recreation, the values of the shapes of the wings were obtained from [1] Boeing's 767 Airplane Characteristics for Airport Planning.

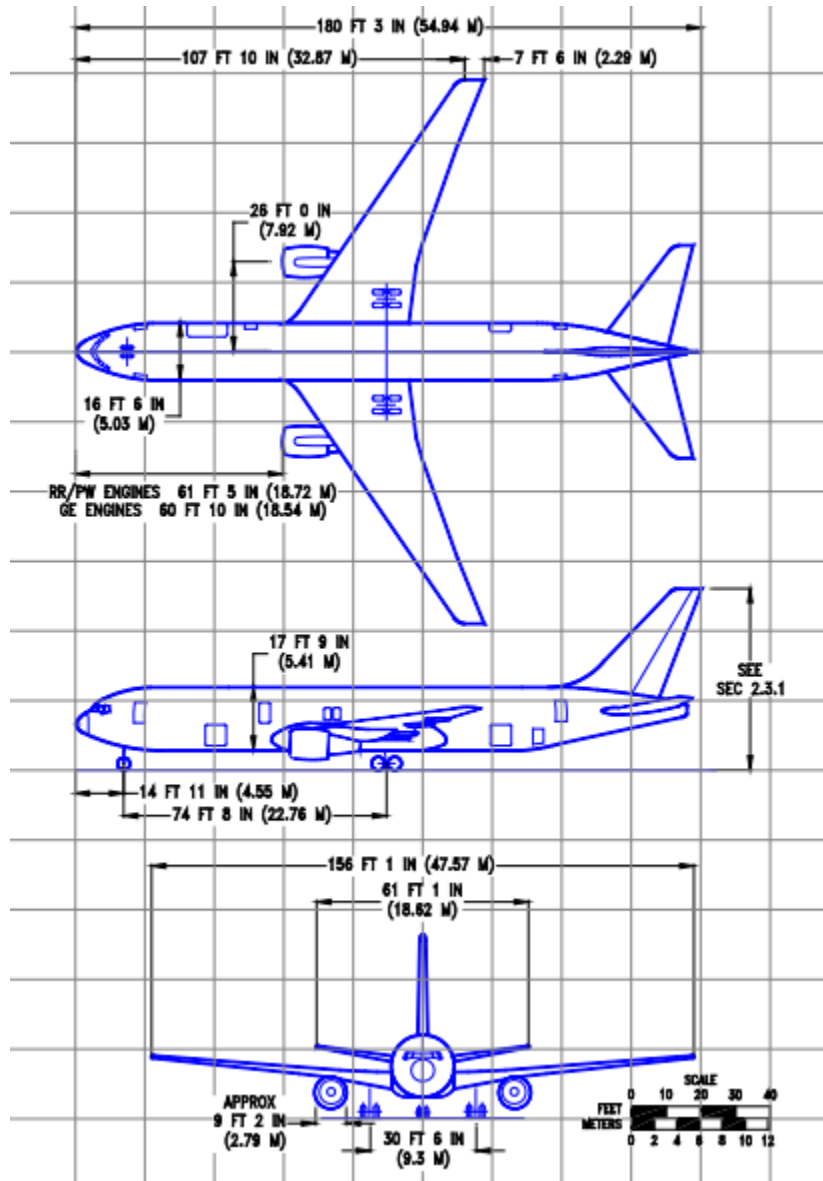


Fig. 1 General Dimensions: Model 767-300, -300ER

Extrapolating from the figure, the following table of values were found for the wing:

Wing Parameters	Value
C_o	11m
C_t	2.29m
$C_{fuselage}$	10m
b_{wing}	47.57m
Δ_{LE}	34°

Fig. 2 Wing Parameters

The first thing to find is the $c(y)$ equation for the wing. The wing is linearly tapered, so the equation for the equation is:

$$c(y) = C_o - \frac{C_o - C_t}{b/2} y \quad (1)$$

Applying extrapolated values:

$$c(y) = 11m - 0.3662y \quad (i)$$

Next characteristic solved was the S_{ref} for the whole wing. Being a trapezoid, the following equation can be used:

$$S_{ref} = \frac{C_o + C_t}{2} * b \quad (2)$$

Applying extrapolated values:

$$S_{ref} = 316.1 m^2 \quad (ii)$$

For the wet surface area of the S_{wet} , Eq. (2) is used, but instead of the root chord, insert the fuselage chord length. Applying the extrapolated values:

$$S_{wet} = 261.4 m^2 \quad (iii)$$

To solve for mean aerodynamic chord $\bar{c}$, λ taper ratio is used, the equations relevant are:

$$\lambda = \frac{C_t}{C_o} \quad (3)$$

$$\bar{c} = \left(\frac{2}{3}\right) * C_o * \frac{1+\lambda+\lambda^2}{1+\lambda} \quad (4)$$

Applying extrapolated values:

$$\bar{c} = 7.5964m \quad (iv)$$

The X_{ac} and Y_{ac} values are also calculated, using the equations:

$$X_{macLE} = \frac{b}{2} \tan(\Delta_{LE}) * \left(\frac{1+2\lambda}{3+3\lambda} \right) \quad (5)$$

$$X_{ac} = X_{macLE} + \frac{\bar{c}}{4} \quad (6)$$

$$Y_{ac} = \frac{b}{2} \left(\frac{1+2\lambda}{3+3\lambda} \right) \quad (7)$$

Applying extrapolated values:

$$\begin{aligned} X_{ac} &= 8.1683m \\ Y_{ac} &= 9.2944m \end{aligned} \quad (iv)$$

The tail parameters are also analyzed, the following table values were found for the tail:

Tail Parameters	Value
C_o	6m
C_t	2m
b_{tail}	18.62m

Fig. 3 Tail Parameters

For $c_t(y)$, Eq. (1) is used with the tail parameters:

$$c_t(y) = 6m - 0.4296y \quad (v)$$

For $\bar{c}$ of the tail, Eq. (3) & (4) are used with tail parameters:

$$\bar{c}_t = 4.33m \quad (v)$$

For S_t , Eq. (2) is used with tail parameters:

$$S_t = 74.48m^2 \quad (v)$$

IV. Center of Gravity Analysis of Various Loading Conditions

The CoG calculations require the weight of various components [1-4]. Since the 767-300 is passenger airline, it can be split it into different categories of masses, which are tabulated below:

Component	Mass (kg)
Fuselage	14,144 kg [4]
Wing (Main)	6,600 kg (Per Wing) [4]
Wing (Tail)	1,880.5 kg (Per Wing)
Engine	23,591 kg (Per Engine) [2-3]
Fuel (Maximum)	50,000 kg [1]
Fuel (Minimum)	20,000 kg [1]
Passengers/Payload (100kg/Passenger)	28,700 kg

These values were obtained from online sources. The mass distributions are not easily found online. Using geometric calculations with scaled drawings [1], we determined the following centroids:

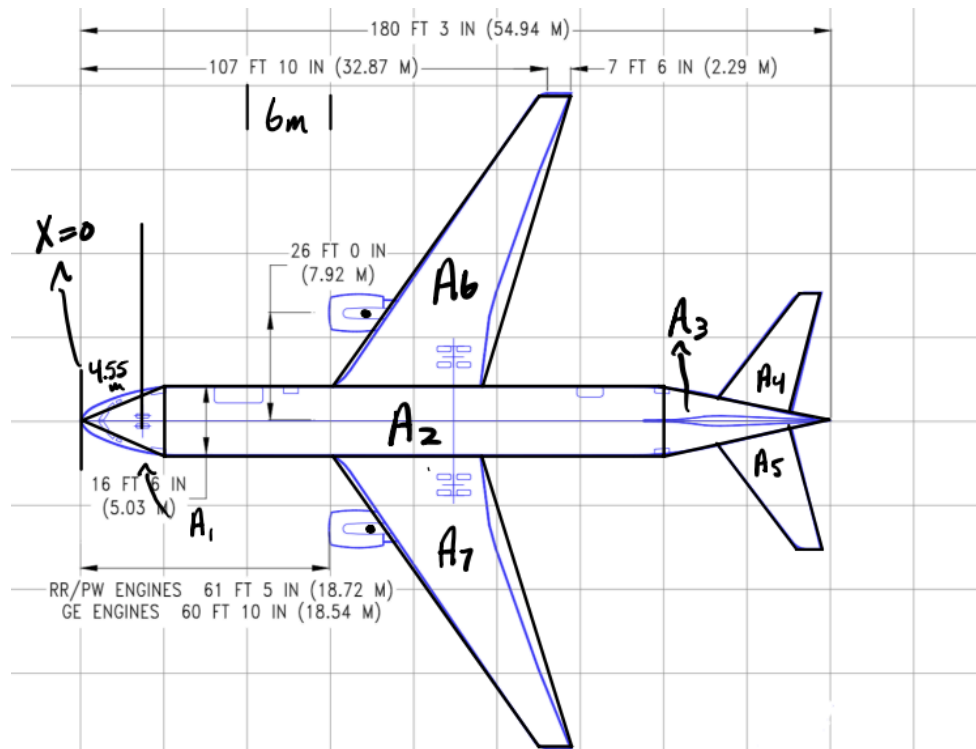


Fig.4 : Scaled drawing of the plane used to determine the centroids [1].

Component	Centroid (Measured from the Nose)
Fuselage (A_1 A_2 A_3)	24 m
Wing (Main) (A_6 A_7)	27.29 m
Wing (Tail) (A_4 A_5)	56 m
Engine	21 m
Fuel (Maximum)	27.29 m
Fuel (Minimum)	27.29 m
Passengers/Payload (100kg/Passenger)	26.855 m

These were found by using the diagrams below and finding their geometric centroids with our datum being at the nose. The fuel centroids are assumed to be at the wing centroid, since the fuel is stored in the wings. The passenger centroid was found by using the seating chart below and assuming a passenger weight of 100 kg. The engine centroid was assumed to be at the center of the engine on the drawing.

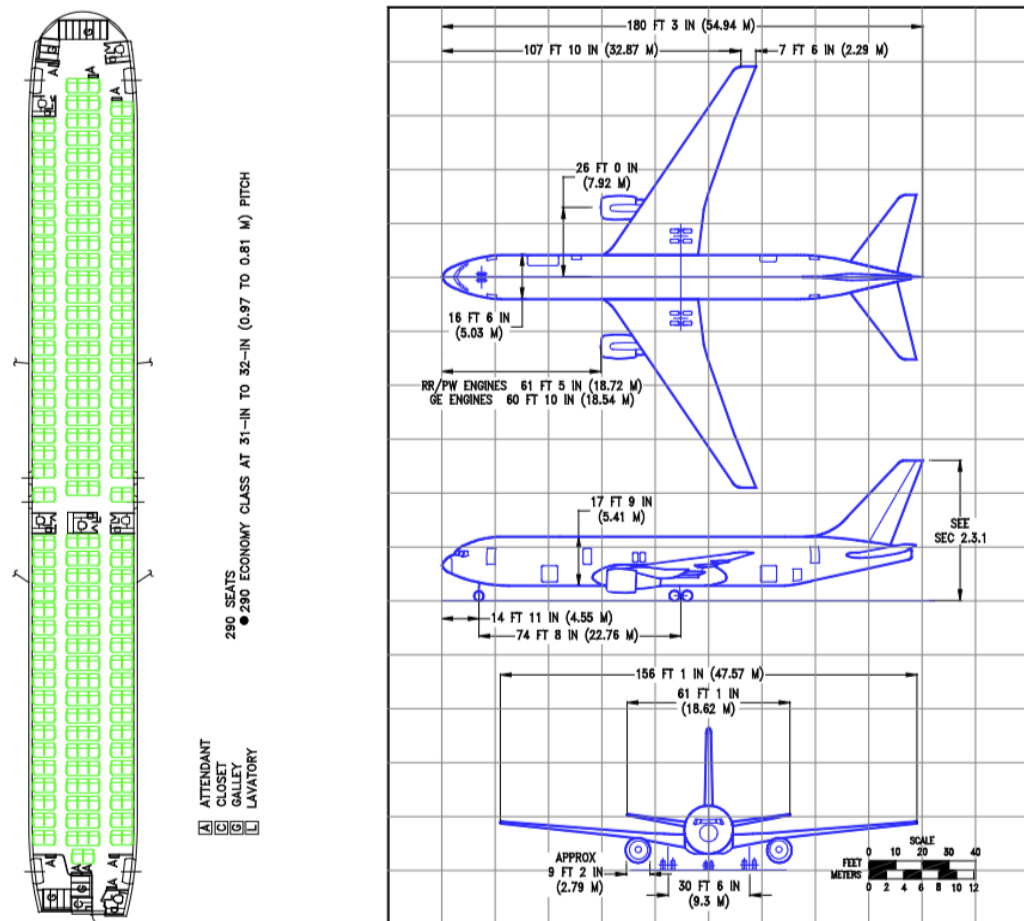


Fig.5 : Left, passenger seating chart. Right, scaled drawing. Both were used to determine centroids [1].
With this data we can determine the different centroids for loading configurations. The equation for the center of gravity for a some given masses is:

$$x_{cent} = \frac{W_1 \cdot x_1 + W_2 \cdot x_2 + W_3 \cdot x_3 + \dots + W_n \cdot x_n}{W_1 + W_2 + W_3 + \dots + W_n}$$

x_i is the object's centroid
 W_i is the object's weight

Using this equation, we get the following:

Loading Configuration	Mass (Kg)	Center of Gravity	% of $\bar{c}$
Empty (Fuselage, Wings, Engines, Minimum Fuel)	98,287 kg	24.89 m	90.7%
Max Fuel (Ferrying Flight) (Fuselage, Wings, Engines, Maximum Fuel)	128,287 kg	25.46 m	98.1%
Max Takeoff (Fuselage, Wings, Engines, Maximum Fuel, Payload)	156,987 kg	25.71 m	101.5%
Landing (Fuselage, Wings, Engines, Minimum Fuel, Payload)	126,987 kg	25.33 m	96.6%

The $x_{cg, AFT}$ and $x_{cg, FWD}$ (calculated in the landing gear section) were $x_{cg, AFT} = 25.991m$ (105.2% $x_{cg, aft}/\bar{c}$) and $x_{cg, FWD} = 24.52 m$ (85.8% $x_{cg, FWD}/\bar{c}$). All of the CoG's calculations all fall within those values, meaning that the plane is safe as long as it is properly loaded. The z_{cg} is 3.95 m, which is calculated in the landing gear section.

The math for this section was done in desmos. A screenshot of the desmos output is below.

Empty Load

add slider: ☐ E ☐ m ☐ p ☐ t ☒ all

$$\frac{x_{centfuse} \cdot W_{fuse} + x_{centwingf} \cdot W_{wingf} \cdot 2 + x_{centwingt} \cdot W_{wingt} \cdot 2 + x_{centeng} \cdot W_{eng} \cdot 2 + x_{centfuel} \cdot W_{minfuel}}{W_{fuse} + W_{wingf} \cdot 2 + W_{wingt} \cdot 2 + W_{eng} \cdot 2 + W_{minfuel}} = 24.8956830507$$

Max Fuel

add slider: ☐ M ☐ a ☐ F ☐ u ☒ all

$$\frac{x_{centfuse} \cdot W_{fuse} + x_{centwingf} \cdot W_{wingf} \cdot 2 + x_{centwingt} \cdot W_{wingt} \cdot 2 + x_{centeng} \cdot W_{eng} \cdot 2 + x_{centfuel} \cdot W_{maxfuel}}{W_{fuse} + W_{wingf} \cdot 2 + W_{wingt} \cdot 2 + W_{eng} \cdot 2 + W_{maxfuel}} = 25.4555956566$$

Max Takeoff

add slider: ☐ M ☐ a ☐ T ☐ k ☒ all

$$\frac{x_{centfuse} \cdot W_{fuse} + x_{centwingf} \cdot W_{wingf} \cdot 2 + x_{centwingt} \cdot W_{wingt} \cdot 2 + x_{centeng} \cdot W_{eng} \cdot 2 + x_{centfuel} \cdot W_{maxfuel} + x_{centpassengers} \cdot W_{passengers}}{W_{fuse} + W_{wingf} \cdot 2 + W_{wingt} \cdot 2 + W_{eng} \cdot 2 + W_{maxfuel} + W_{passengers}} = 25.7114315198$$

Landing

add slider: ☐ L ☐ a ☐ n ☐ d ☒ all

$$\frac{x_{centfuse} \cdot W_{fuse} + x_{centwingf} \cdot W_{wingf} \cdot 2 + x_{centwingt} \cdot W_{wingt} \cdot 2 + x_{centeng} \cdot W_{eng} \cdot 2 + x_{centfuel} \cdot W_{minfuel} + x_{centpassengers} \cdot W_{passengers}}{W_{fuse} + W_{wingf} \cdot 2 + W_{wingt} \cdot 2 + W_{eng} \cdot 2 + W_{minfuel} + W_{passengers}} = 25.3385031539$$

V. Stability Analysis Using DATCOM

Below are the MATLAB visualizations of the DATCOM input code. The NACA profile for the Boeing 767 wing was unavailable, so that of the Mig 17 was used instead. The vertical and horizontal stabilizers are those of the Boeing 767.

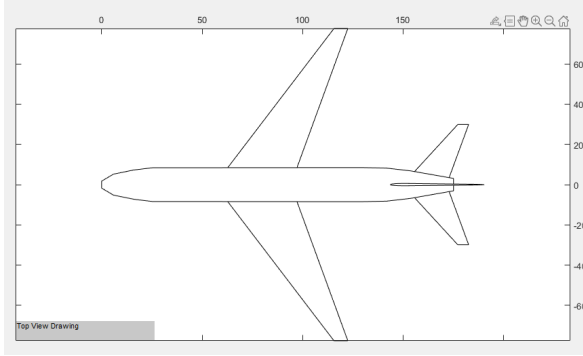


Figure 6: Top view MATLAB drawing of the Boeing 747 based on DATCOM inputs.

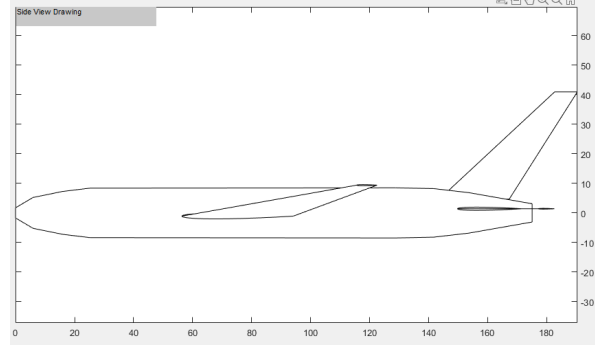


Figure 7: Sideview MATLAB drawing of the Boeing 747 based on DATCOM inputs.

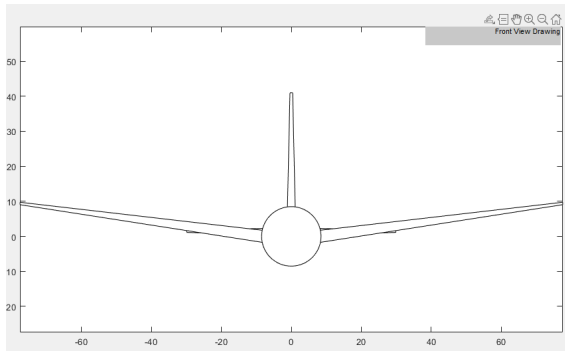


Figure 8: Front view MATLAB drawing of the Boeing 747 based on DATCOM inputs.

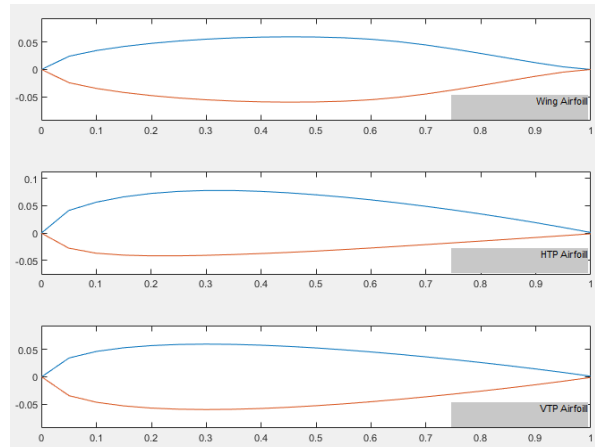


Figure 9: Airfoil MATLAB drawing of the wing, horizontal stabilizer, and vertical stabilizer based on DATCOM inputs.

Upon inputting the three angles of attack, -2° , 2° , and 8° and the two speeds, 0.15 and 0.6 Mach, as shown in the figure below, DATCOM provided a stability analysis of the plane and its components for each of the listed conditions. As shown in Figures A1 and A2, the C_{MA} longitudinal stability derivatives for the three angles of attack at 0.15 and 0.6 Mach are all negative, which means the plane remains stable.

```

CASEID --
$FLTCN NMACH=2.0,MACH(1)=0.15,0.6,NALPHA=3.0,ALSCHD(1)=-2.0,2.0,
8.0,NALT=1.,ALT(1)=1.0,
WT=345000.0,LOOP=2.$
$SYNTHS XCG=92.2,ZCG=0.0,XW=56.3,ZW=-1.1,ALIW=1.0,XH=149.7,
ZH=1.4,ALIH=0.0,XV=143.6,ZV=4.67$
$OPTINS SREF=3050.0$
$BODY NX=8.0,
X(1)=0.0,5.92,15.98,25.19,128.74,141.71,153.62,175.06,
S(1)=9.40,85.73,165.79,220.28,226.54,212.19,147.40,30.,$
NACA-W-6-66-012
$WGPLNF CHRDP=6.9,SSPNOP=53.2,SSPNE=69.4,SSPN=77.6,
CHRD BP=24.3,CHRD R=37.8,SAVSI=33.6,SAVSO=33.6,CHSTAT=0.25,
TWISTA=-3.,DHDADI=7.7,DHDADO=6.,TYPE=1.0$
NACA-H-4-2412-25
$HTPLNF CHRDP=5.4,SSPNE=27.9,SSPN=29.9,CHRD R=21.9,SAVSI=38.,
CHSTAT=.25,TYPE=1.0$
NACA-V-4-0012-25
$VTPLNF CHRDP=7.7,SSPNE=31.4,SSPN=36.3,CHRD R=23.7,SAVSI=44.0,
CHSTAT=0.25,TYPE=1.0$
DIM FT
BUILD
PLOT
NEXT CASE

```

Figure 10: DATCOM code for Boeing 767.

When the plane is undergoing a 0° sideslip, it remains stable. As shown in Figures A1 and A2, the C_{NB} stability derivative is positive, which results in the plane being directionally stable. It is also laterally stable, as the C_{LB} is negative. Considering the C_{NB} remains positive and C_{LB} remains negative through all angles of attack, this plane will likely resist sideslip; therefore, it is expected to remain stable at 10° sideslip as well.

To calculate the neutral point, the equation below is used.

$$\frac{x_{ac}}{\bar{c}} + v_{HT} * \left(1 - \frac{\partial \epsilon}{\partial \alpha}\right) * \left(\frac{\frac{\partial C_{Lt}}{\partial \alpha}}{\frac{\partial C_{LW}}{\partial \alpha}}\right) = \frac{26.77}{24.9} + 0.69 \cdot (1 - 0.375) \left(\frac{0.052}{0.11}\right)$$

The second ratio is pulled from Figure A2 at an angle of attack of 2° , while the third and fourth ratios are found in Figures A3 and A4.

By using the X_{ac} with respect to the mean aerodynamic chord, the first ratio amounts to 1.075. To find V_{HT} , the equation below was used.

$$v_{HT} = \frac{S_t * l_T}{S_w * \bar{c}} = \frac{74.5 \cdot 18.3}{261 \cdot 7.6}$$

With values in the equation being calculated in previous sections, this ratio amounts to 0.69. Plugging all of these values into the neutral point equation, this value is found to be **1.27**.

The static margin can be calculated with the equation below. Using the X_{cg} in relation to the mean aerodynamic chord, 6.26 m, plugging it all in results with the answer **0.45**.

$$\frac{h}{\bar{c}} = -\frac{(x_{cg} - x_{ac})}{\bar{c}} + v_{HT} * \left(1 - \frac{\partial \epsilon}{\partial \alpha}\right) * \left(\frac{\frac{\partial C_{Lt}}{\partial \alpha}}{\frac{\partial C_{Lw}}{\partial \alpha}}\right) - \frac{(6.26 - 8.16)}{7.6} + 0.69 \cdot (1 - 0.375) \left(\frac{0.052}{0.11}\right)$$

To calculate the phugoid damping frequency and damping ratio, the following equations were used.

$$\omega_{ph} = \sqrt{2} * \frac{g}{V_{\infty}} \quad \zeta_{ph} = \frac{1}{\sqrt{2}} \frac{C_D}{C_L}$$

For Mach 0.6, or 205 m/s, the phugoid damping frequency is **0.067 1/s**, and the damping ratio is **0.059**. The values for the C_D/C_L ratio were pulled from Figure A2.

$$\sqrt{2} \cdot \frac{9.81}{205} \quad \frac{1}{\sqrt{2}} \cdot \frac{.016}{.193}$$

For Mach 0.15, or 51.45 m/s, the phugoid damping frequency is **0.27 1/s**, and the damping ratio is **0.067**. The values for the C_D/C_L ratio were pulled from Figure A1.

$$\sqrt{2} \cdot \frac{9.81}{51.45} \quad \frac{1}{\sqrt{2}} \cdot \frac{.016}{.177}$$

VI. Landing Gear Calculations

In order to begin landing gear calculations, diagrams and information pertaining to the landing gear were obtained from [1] Boeing's 767 Airplane Characteristics for Airport Planning.

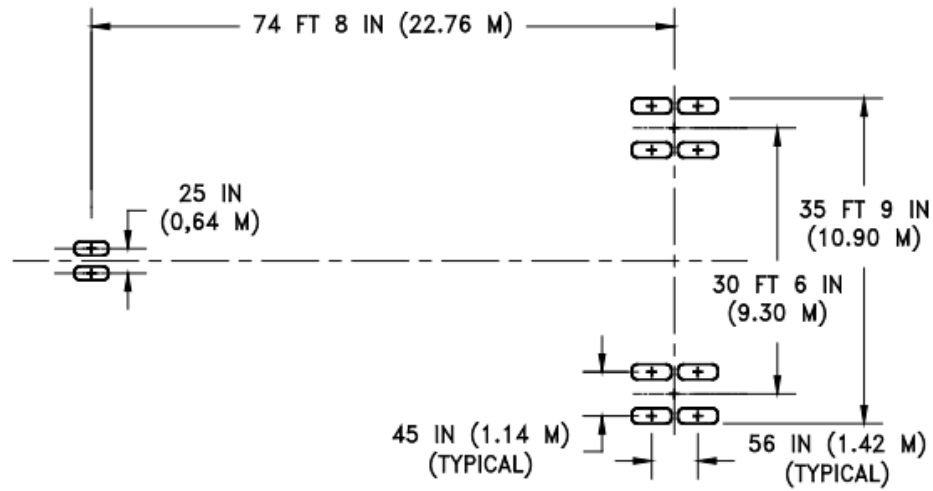


Fig. 11 General Dimensions for Wheels: Model 767-300, -300ER

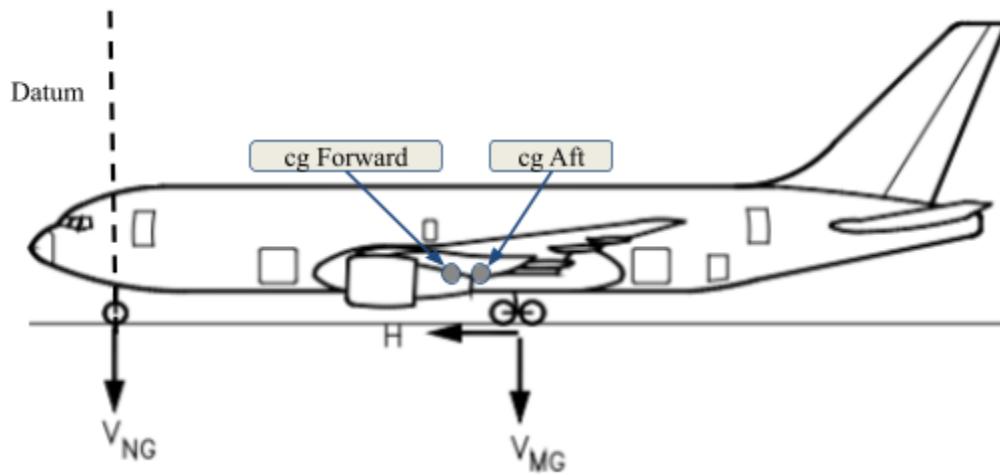


Fig. 12 Location of Landing Gear Loads

Extrapolating from the above figures and from information provided by [1] Boeing's 767 Airplane Characteristics for Airport Planning:

W [kg]	163,746
V_{MG} [kg / strut]	77,130
# of struts	2
V_{NG} [kg]	20,455
B [m]	22.76
X_{MLG} [m]	22.76
z_{CG} [m]	3.947
A [m]	4.65

Fig. 13 Landing Gear Loading Parameters

From the above figure, the locations of airplanes X_{cg} and Z_{cg} for forward and aft conditions could be found utilizing Eq. (8) and Eq. (9).

$$\max L_{MG} = W * \frac{N_a}{B} \quad (8)$$

$$\max L_{NG} = W * \frac{M_f}{B} \quad (9)$$

Rearranging to solve for the center of gravities:

$$x_{cg \text{ Aft}} = N_a = \frac{L_{MG} * B}{W} \quad (10)$$

$$M_f = \frac{L_{NG} * B}{W} \quad (11)$$

$$x_{cg \text{ Forward}} = B - M_f \quad (12)$$

Using extrapolated value:

$$X_{cg \text{ Aft}} = 21.441 \text{ [m]}$$

$$x_{cg \text{ Forward}} = 19.917 \text{ [m]}$$

To solve for the max main landing gear load and max nose landing gear load Eq. (8) and Eq(9) can be utilized again resulting in:

$$\max L_{MG} = 1,513,187.707 [N]$$

$$\max L_{NG} = 200,440.644 [N]$$

Solving for minimum nose landing gear load, Eq. (13) was utilized.

$$\min. L_{NG} = W \frac{M_a}{B} \quad (13)$$

$$\min. L_{NG} = 93,162.553 [N]$$

To solve for maximum breaking landing gear load, Eq. (14) was utilized knowing $a = [3\frac{m}{s^2}]$, $g = 9.81[\frac{m}{s^2}]$ and $H = z_{cg}$.

$$\max \text{ breaking } L_{NG} = \frac{a}{g} * \frac{H*W}{B} \quad (14)$$

$$\max \text{ breaking } L_{NG} = 85,189.648 [N]$$

Tip back angle could then be calculated using Eq. (15) and found values.

$$\tan \Theta = \frac{x_{MLG} - x_{cg Aft}}{z_{cg}} \quad (15)$$

$$\Theta = \tan^{-1} \left(\frac{x_{MLG} - x_{cg Aft}}{z_{cg}} \right)$$

$$\Theta = 18.48 \text{ degrees}$$

For turnover angle, Eq. (16), Eq. (17), and Eq. (18) were all used.

$$\alpha = \tan^{-1}(A/B) \quad (16)$$

$$\alpha = 11.547 \text{ degrees}$$

$$y = (D + X)\sin(\alpha) \quad (17)$$

$$y = D * \sin(\alpha) = x_{CG Aft} * \sin(\alpha)$$

$$y = 4.292 [m]$$

$$\Phi = \tan^{-1}(H/Y) \quad (18)$$

$$\Phi = 42.602 \text{ degrees}$$

For calculating tire size there are two methods that can be utilized. The first method utilizes Fig. 14 for a transport classification of airplanes.

	Diameter		Width	
	A	B	A	B
<i>British units: Main wheels diameter or width (in.) = A W_w^B</i>				
General aviation	1.51	0.349	0.7150	0.312
Business twin	2.69	0.251	1.170	0.216
Transport/bomber	1.63	0.315	0.1043	0.480
Jet fighter/trainer	1.59	0.302	0.0980	0.467
<i>Metric units: Main wheels diameter or width (cm) = A W_w^B</i>				
General aviation	5.1	0.349	2.3	0.312
Business twin	8.3	0.251	3.5	0.216
Transport/bomber	5.3	0.315	0.39	0.480
Jet fighter/trainer	5.1	0.302	0.36	0.467

Fig. 14 Main wheel diameter and width conversion table

number of MLG wheels	8
W_{WMLG} [kg]	19432.50
number of NLG wheels	2
W_{WNLG} [kg]	10227.5

Fig. 15 Wheel Parameters

$$Diameter\ MLG = 5.3(W_{WMLG})^{0.315} \quad (19)$$

$$Diameter\ MLG = 118.895 [cm]$$

$$Width\ MLG = 0.39(W_{WMLG})^{0.480} \quad (20)$$

$$Width\ MLG = 44.623 [cm]$$

$$Diameter\ NLG = 5.3(W_{WNLG})^{0.315} \quad (21)$$

$$Diameter\ NLG = 97.130 [cm]$$

$$Width\ NLG = 0.39(W_{WNLG})^{0.480} \quad (22)$$

$$Width\ NLG = 32.791 [cm]$$

Actual tire size, according to [1] Boeing's 767 Airplane Characteristics for Airport Planning is:

MLG Tire Size [inches]	50x20 R22 32PR
NLG Tire Size [inches]	H37x14-15 22PR

The calculated values for wheel size falls in line with the actual tire sizing for both main and nose landing gears. The second method utilized braking energy dissipation utilizing Eq. (23) and Fig. 16.

$$KE = \frac{1*W}{2*g} * V_{Stall}^2 \quad (23)$$

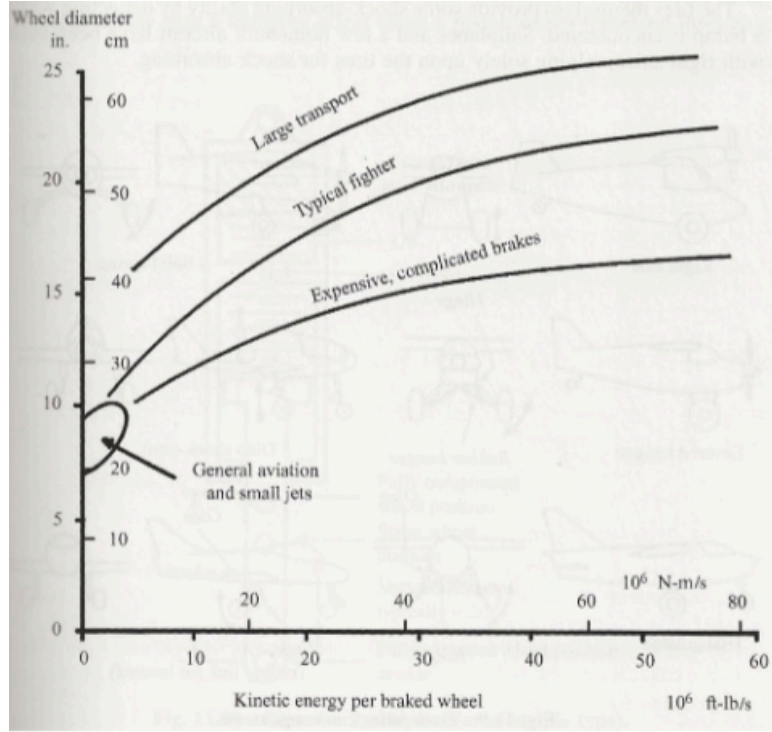


Fig.16 Kinetic Energy per Wheel to Wheel Diameter Graph

$$KE_{MLG} = 44.805 * 10^6 [Nm]$$

$$Diameter_{MLG} = 60.5 [cm]$$

$$KE_{NLG} = 23.581 * 10^6 [Nm]$$

$$Diameter_{NLG} = 56 [cm]$$

Comparing these values to the actual there is some variation in overall sizing.

VI. Conclusion

The Boeing 767 -300 is an engineering marvel that pursues to push the limits of commercial flying. This paper has undertaken the endeavor of analyzing the many aspects of this plane from wing airfoil calculations to a DATCOM analysis of the fuselage to an analysis of loading conditions and an analysis of the landing gear.

Appendix A

AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP WING-BODY-VERTICAL TAIL-HORIZONTAL TAIL CONFIGURATION											

MACH		ALTITUDE		FLIGHT CONDITIONS		TEMPERATURE		REYNOLDS		REFERENCE DIMENSIONS	
NUMBER											
	FT	FT/SEC	LB/FT**2		DEG R		1/FT	REF. AREA	REFERENCE LENGTH	LAT. LAT.	MOMENT REF. CENTER
								FT**2	FT	FT	FT
0 0.150	1.00	167.45	2.1161E+03		518.666		1.0602E+06	3050.000	24.052	155.200	92.200 0.000
0 0	-----										
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB	CLB
0											
-2.0	0.015	-0.147	-0.0643	-0.147	0.010	0.436	8.094E-02	-7.745E-03	-1.195E-02	2.209E-03	-2.592E-03
2.0	0.016	0.177	-0.0938	0.178	0.010	-0.528	8.172E-02	-9.611E-03			-2.928E-03
8.0	0.041	0.674	-0.1715	0.673	-0.053	-0.255	8.394E-02	-1.629E-02			-3.439E-03
0 0	-----										
			ALPHA	Q/QINF	EPSLON	D(EPSLON)/D(ALPHA)					
			-2.0	1.000	-0.782	0.343					
			2.0	0.982	0.676	0.356					
			8.0	1.000	2.732	0.343					
0*** VEHICLE WEIGHT = 345000.00 LB.											
0*** LEVEL FLIGHT LIFT COEFFICIENT = 3.39386											

Figure 1: DATCOM output table for the entire Boeing 767 at 0.15 Mach.

1 AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP WING-BODY-VERTICAL TAIL-HORIZONTAL TAIL CONFIGURATION ---											
----- FLIGHT CONDITIONS -----						----- REFERENCE DIMENSIONS -----					
MACH NUMBER	ALTITUDE	VELOCITY	PRESSURE		TEMPERATURE	REYNOLDS NUMBER	REF. AREA	REFERENCE LONG.	LENGTH LAT.	MOMENT HORIZ	REF. CENTER VERT
	FT	FT/SEC	LB/FT**2		DEG R	1/FT	FT**2	FT	FT	FT	FT
0 0.600	1.00	669.79	2.1161E+03		518.666	4.2408E+06	3050.000	24.052	155.200	92.200	0.000
----- DERIVATIVE (PER DEGREE) -----											
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB	CLB
0											
-2.0	0.015	-0.147	-0.0694	-0.148	0.009	0.470	8.507E-02	-8.543E-03	-1.266E-02	2.216E-03	-2.778E-03
2.0	0.016	0.193	-0.1007	0.193	0.009	-0.521	8.573E-02	-1.002E-02			-3.221E-03
8.0	0.041	0.714	-0.1805	0.713	-0.059	-0.253	8.791E-02	-1.659E-02			-3.892E-03
0			ALPHA		Q/QINF	EPSLON	D(EPSLON)/D(ALPHA)				
0											
			-2.0		1.000	-0.826	0.357				
			2.0		0.989	0.722	0.375				
			8.0		1.000	2.861	0.357				
0*** VEHICLE WEIGHT = 345000.00 LB.											
0*** LEVEL FLIGHT LIFT COEFFICIENT = 0.21212											

Figure A2: DATCOM output table for the entire Boeing 767 at 0.6 Mach.

1	AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP HORIZONTAL TAIL CONFIGURATION ---											
----- FLIGHT CONDITIONS -----												
MACH	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS	REFERENCE DIMENSIONS						
NUMBER					NUMBER	REF.	REFERENCE	LENGTH	MOMENT	REF.	CENTER	
	FT	FT/SEC	LB/FT**2	DEG R	1/FT	AREA	LONG.	LAT.	HORIZ	VERT		
						FT**2	FT	FT	FT	FT	FT	
0 0.150	1.00	167.45	2.1161E+03	518.666	1.0602E+06	3050.000	24.052	155.200	92.200	0.000		
0	----- DERIVATIVE (PER DEGREE) -----											
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB	CLB	
0												
-2.0	0.002	-0.003	0.0048	-0.003	0.002	-1.843	1.299E-02	-3.965E-02	3.231E-09	9.585E-09	3.484E-06	
2.0	0.003	0.052	-0.1642	0.053	0.001	-3.124	1.453E-02	-4.487E-02	1.346E-06	4.021E-06	-7.113E-05	
8.0	0.012	0.147	-0.4569	0.147	-0.008	-3.111	1.685E-02	-5.269E-02	1.050E-05	3.139E-05	-1.987E-04	

Figure A3: DATCOM output for the horizontal tail alone of the Boeing 767.

1	AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP WING ALONE CONFIGURATION --										
	FLIGHT CONDITIONS					REFERENCE DIMENSIONS					
	MACH	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS	REF.	REFERENCE	MOMENT	REF. CENTER	
	NUMBER					NUMBER	AREA	LENGTH			
		FT	FT/SEC	LB/FT**2	DEG R	1/FT	FT**2	LONG.	LAT.	HORIZ	VERT
0	0.150	1.00	167.45	2.1161E+03	518.666	1.0602E+06	3050.000	24.052	155.200	92.200	0.000
0	DERIVATIVE (PER DEGREE)										
0	ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB
0											CLB
	-2.0	0.007	-0.125	0.0015	-0.126	0.002	-0.012	5.881E-02	6.637E-03	-7.704E-04	1.283E-05
	2.0	0.006	0.110	0.0281	0.110	0.003	0.255	5.814E-02	6.605E-03	-7.709E-04	9.828E-06
	8.0	0.020	0.453	0.0674	0.451	-0.043	0.149	5.624E-02	6.509E-03	-7.426E-04	1.674E-04
											-2.537E-03

Figure A4: DATCOM output for the wing alone of the Boeing 767.

References

- [1] Boeing Commercial Airplanes, “767 Characteristics for Airport Planning” Boeing D6-58328, 2021.
<https://upcommons.upc.edu/bitstream/handle/2117/106436/memoria.pdf;jsessionid=EBDA3CBBF2FE6E385DE1A3E72CB7A53B?sequence=1>
- [2] “Boeing 767-300ER Tech Info :: Quick Facts.” Retrieved 9 March 2025.
http://hibdz.skydiving.co.uk/757/767_tech/main/quick_facts.htm
- [3] “Boeing 767-300 - Specifications - Technical Data / Description.” Retrieved 9 March 2025.
https://www.flugzeuginfo.net/acdata_php/acdata_7673_en.php
- [4] Times, G. B., “How Much Does a Boeing 767 Weigh?,” GB Times, Jul 30 2024. Retrieved 9 March 2025.
<https://gbtimes.com/how-much-does-a-boeing-767-weigh/>